Mahmoud Alyosify

Machine Learning Engineer

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SUMMARY

Machine learning engineer who takes models **from raw data to a working service**: data preparation, training, fair evaluation and deployment with FastAPI, ONNX, Docker and AWS. Projects span computer vision (84.30% top-1 accuracy with self-supervised learning), risk prediction on imbalanced data, deep reinforcement learning and LLM fine-tuning. MSc in Artificial Intelligence, Queen's University, Canada (GPA 4.30 / 4.30); currently an AI Engineer Intern at Injaz Tech.

SKILLS

Machine learning: PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, deep learning, CNNs, computer vision, NLP

Workflow: data cleaning, feature engineering, model evaluation, hyperparameter tuning (Optuna), imbalanced data, SHAP

LLMs: RAG, LangChain, LangGraph, fine-tuning (LoRA, QLoRA), Hugging Face Transformers

Deployment: FastAPI, ONNX, FAISS, Docker, AWS, GitHub Actions, pytest, Git

Data: Python, SQL, Pandas, NumPy, PySpark, Matplotlib

EXPERIENCE

AI Engineer Intern — Oct 2026 – Present
Injaz Tech (Injaz Group, formerly Fingerprint Consultancy) — Maadi, Cairo

- Building a RAG assistant that answers employees' questions from company documents, using **LangChain, LangGraph, FAISS** and **FastAPI**.
- Raised top-1 retrieval accuracy from **75% to 94%** and cut wrong refusals from 5 in 20 to 1 in 20.
- Covered the service with **136 automated tests** (pytest) and an evaluation check that compares each new version with the last.

Part-Time Machine Learning Instructor — Jul 2025 – Present
National Telecommunication Institute (NTI) — Egypt

- Teach **90-hour hands-on machine learning bootcamps**: regression, SVM, KNN, Naive Bayes, PCA and neural networks, in Python labs.

AI in Cybersecurity Intern — Jun 2025 – Jul 2025
e& Egypt (on-the-job training via NTI) — Cairo

- Applied AI tools to threat intelligence, digital forensics and enterprise security analysis.

Systems Engineer (Military Service) — Feb 2024 – Mar 2025
National Company for Roads Building & Development — Cairo

- Supported the backend systems of Egypt's national toll network, used by **more than half of the country's vehicles** across 70%+ of national roads.
- Co-developed an internal data-management and reporting application in **C#, .NET, Entity Framework, SQL Server** and Crystal Reports.

Online Instructor — 2020 – Present
Udemy & YouTube (Arabic courses) — Remote

- Created Arabic courses on algorithms, discrete mathematics, probability and statistics: **10,000+ learners, 700K+ views**.

PROJECTS

SimCLR-Vision-SSL — self-supervised image model — 2026
PyTorch, SimCLR, SupCon, ONNX, FAISS

- Trained an image model without labels (SimCLR), tested **42 augmentation settings** and reached **84.30% top-1 accuracy** with a semi-supervised extension (SupCon).
- Exported the model to **ONNX** and served image search through a **FAISS** index, ready to deploy.

SCRAP — satellite collision risk prediction 2026
Python, XGBoost, LightGBM, Optuna, SHAP

- Predicts satellite collision risk **48+ hours** ahead: **97.31% recall** on high-risk events. Interactive 3D demo online.

Horus-OSINT — fine-tuned LLM on AWS 2026
Python, Llama-3-8B, QLoRA, PySpark, AWS (EC2, S3, EMR), Terraform

- Fine-tuned **Meta-Llama-3-8B** on **159,826** global event and terrorism records (GTD, GDELT) and deployed it on **AWS** as an assistant that answers questions in plain language.

RL-Cloud-Autoscaler — deep RL for cloud scaling 2026
Python, Gymnasium, PPO, DQN

- Built a cloud simulation (Gymnasium) and trained **PPO** and **DQN** agents to balance cost and latency, compared with standard threshold-based autoscaling.

MinimalLM — Master's thesis, published on arXiv 2026
Python, PyTorch, Hugging Face Transformers, NVIDIA A40

- MSc thesis: a training-free method that makes open LLMs generate **47.6% fewer tokens** and answer 19–68% faster, with no answer cut off.

Also: AudioShield (deepfake audio detection) · HORUS Sentinel (multi-agent OSINT) · Vitalism Solution (vital signs from video, Imagine Cup 2023 semi-finalist) · Diabetes Onset Prediction.

PUBLICATIONS

Say Less, Mean Better: Training-Free Verbosity Control for Language Models
M. Alyosify, M. Imbabi, E. Elkhamisi, R. Ferdousi · arXiv:2611.04107 (cs.CL) · November 2026

Beyond Truncation? Entailment-Licensed Elision and the Limits of Semantic Preservation in LLM Decoding
M. Alyosify, M. Imbabi, E. Elkhamisi · arXiv:2611.08975 (cs.CL) · November 2026

EDUCATION

MSc in Artificial Intelligence Sep 2025 – Oct 2026
Queen's University Kingston, Canada

- GPA **4.30 / 4.30** · Completed October 2026 · **Digilians Presidential Scholarship** (competitive national AI fellowship)
- Thesis: MinimalLM — shorter LLM answers that stay complete (supervisor: Dr. Rahatara Ferdousi); published on arXiv
- Coursework: Large Language Models, Generative AI, Deep Learning, Reinforcement Learning, Cloud Computing, End-to-End ML Pipelines

BSc in Computer & Information Science (Bioinformatics) Sep 2019 – Jul 2023
Assiut University Assiut, Egypt

- GPA 3.53 / 4.00 (Very Good with Honours) · Graduation project graded A+

CERTIFICATIONS

- AWS Certified Cloud Practitioner (Amazon Web Services)
- IBM AI Developer Professional Certificate (IBM, 2026)
- Deep Learning Specialization (DeepLearning.AI, 2026)
- Machine Learning Specialization (DeepLearning.AI & Stanford, 2026)
- Machine Learning Internship, ITIDA / Egypt Makes Electronics (200 hours, 2023)

AWARDS & LEADERSHIP

- 1st Place, Smart Cities Hackathon 2022 (77 university teams)
- Semi-Finalist, Microsoft Imagine Cup 2023
- Finalist, ISEIC 2023 (international research competition)
- Secretary, Higher Scientific & Technological Committee, Student Union, Assiut University

LANGUAGES

Arabic (native) · English (professional working proficiency)